

COMPUTING FUNDAMENTALS: HARDWARE, SOFTWARE, NETWORKS AND CLOUD

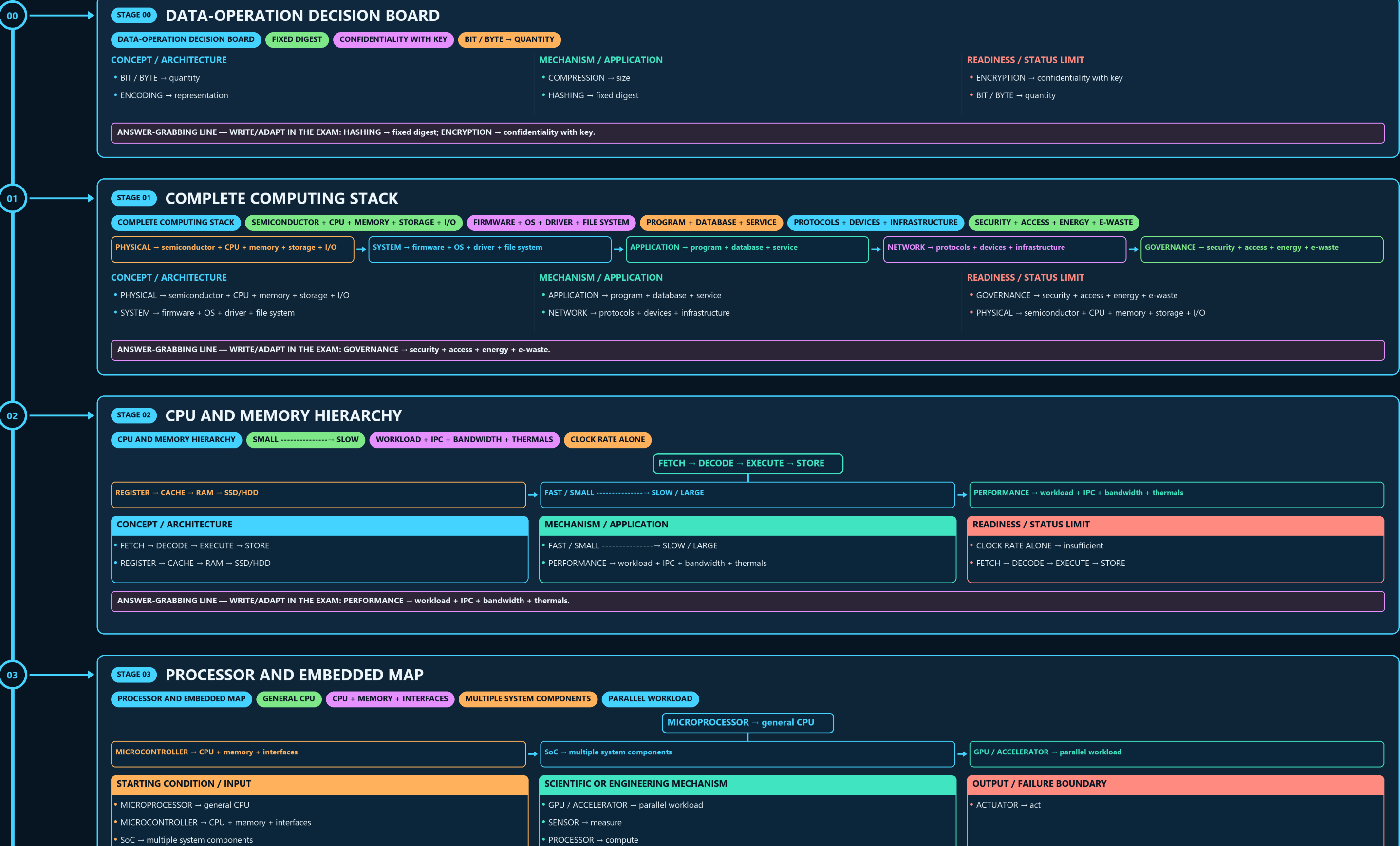
DATA-OPERATION DECISION BOARD → COMPLETE COMPUTING STACK → CPU AND MEMORY HIERARCHY → PROCESSOR AND EMBEDDED MAP → PROGRAM AND EXECUTION LADDER → NETWORK DEVICE AND LAYER

Read the thick cyan rail from Stage 00 to the final core synthesis. Each card uses a concept-appropriate structure; the grey E card is optional enrichment only and never repairs missing core content.

PRIMARY CORE FLOW SUBORDINATE EXTRA APPROVAL / REVIEW

Approval: FALSE • Pending user review • source generation g2 and all prior artifacts unchanged

CORE BEFORE EXTRA • prior artefacts unchanged • exact same master drives poster and tiles



03

STAGE 03 PROCESSOR AND EMBEDDED MAP

PROCESSOR AND EMBEDDED MAP

GENERAL CPU

CPU + MEMORY + INTERFACES

MULTIPLE SYSTEM COMPONENTS

PARALLEL WORKLOAD

MICROPROCESSOR → general CPU

MICROCONTROLLER → CPU + memory + interfaces

SoC → multiple system components

GPU / ACCELERATOR → parallel workload

STARTING CONDITION / INPUT

- MICROPROCESSOR → general CPU
- MICROCONTROLLER → CPU + memory + interfaces
- SoC → multiple system components

SCIENTIFIC OR ENGINEERING MECHANISM

- GPU / ACCELERATOR → parallel workload
- SENSOR → measure
- PROCESSOR → compute

OUTPUT / FAILURE BOUNDARY

- ACTUATOR → act

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: SENSOR → measure | PROCESSOR → compute | ACTUATOR → act.

04

STAGE 04 PROGRAM AND EXECUTION LADDER

PROGRAM AND EXECUTION LADDER

OVERLAPPING PROGRESS

SIMULTANEOUS EXECUTION

COORDINATION FAILURES

SOURCE → compiler / interpreter / assembler

PROGRAM → PROCESS → THREAD

CONCURRENCY → overlapping progress

PARALLELISM → simultaneous execution

STARTING CONDITION / INPUT

- SOURCE → compiler / interpreter / assembler
- PROGRAM → PROCESS → THREAD

SCIENTIFIC OR ENGINEERING MECHANISM

- CONCURRENCY → overlapping progress
- PARALLELISM → simultaneous execution

OUTPUT / FAILURE BOUNDARY

- RACE / DEADLOCK → coordination failures
- SOURCE → compiler / interpreter / assembler

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: PARALLELISM → simultaneous execution; RACE / DEADLOCK → coordination failures.

05

STAGE 05 NETWORK DEVICE AND LAYER MAP

NETWORK DEVICE AND LAYER MAP

IP + ROUTER

WI-FI + SWITCH

APPLICATION → HTTP

APPLICATION → HTTP

TRANSPORT → TCP

INTERNET → IP + router

LINK → Ethernet

CONCEPT / ARCHITECTURE

- APPLICATION → HTTP
- TRANSPORT → TCP

MECHANISM / APPLICATION

- INTERNET → IP + router
- LINK → Ethernet

READINESS / STATUS LIMIT

- Wi-Fi + switch
- PHYSICAL → electrical

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: LINK → Ethernet | Wi-Fi + switch; PHYSICAL → electrical | optical | radio.

06

STAGE 06 INTERNET AND WEB REQUEST FLOW

INTERNET AND WEB REQUEST FLOW

DOMAIN →DNS→ IP ADDRESS

BROWSER →HTTP/HTTPS→ WEB SERVICE

BROWSER-HELD STATE

PROTECTED TRANSPORT, NOT TRUTHFUL CONTENT

DOMAIN →DNS→ IP ADDRESS

BROWSER →HTTP/HTTPS→ WEB SERVICE

TCP/UDP → transport

IP → routing

COOKIE → browser-held state / identifier

STARTING CONDITION / INPUT

- DOMAIN →DNS→ IP ADDRESS
- BROWSER →HTTP/HTTPS→ WEB SERVICE

SCIENTIFIC OR ENGINEERING MECHANISM

- TCP/UDP → transport
- IP → routing

OUTPUT / FAILURE BOUNDARY

- COOKIE → browser-held state / identifier
- HTTPS → protected transport, not truthful content

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: COOKIE → browser-held state / identifier; HTTPS → protected transport, not truthful content.

06

STAGE 06 INTERNET AND WEB REQUEST FLOW

INTERNET AND WEB REQUEST FLOW DOMAIN --DNS--> IP ADDRESS BROWSER --HTTP/HTTPS--> WEB SERVICE BROWSER-HELD STATE PROTECTED TRANSPORT, NOT TRUTHFUL CONTENT



STARTING CONDITION / INPUT

- DOMAIN --DNS--> IP ADDRESS
- BROWSER --HTTP/HTTPS--> WEB SERVICE

SCIENTIFIC OR ENGINEERING MECHANISM

- TCP/UDP -> transport
- IP -> routing

OUTPUT / FAILURE BOUNDARY

- COOKIE -> browser-held state / identifier
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07

STAGE 07 PERFORMANCE AND WIRELESS MATRIX

PERFORMANCE AND WIRELESS MATRIX ACHIEVED RATE DELAY VARIATION PACKET CELLULAR VOICE OVER LTE DISTINCT ROLES WIRELESS LABEL NO UNIVERSAL RANGE OR WALL PENETRATION

BANDWIDTH -> capacity	THROUGHPUT -> achieved rate
LATENCY -> delay	JITTER -> delay variation
LTE -> packet cellular	VoLTE -> voice over LTE

SYSTEM / DEVICE / INSTITUTION

- BANDWIDTH -> capacity
- THROUGHPUT -> achieved rate
- LATENCY -> delay

DECISIVE TECHNICAL DISTINCTION

- JITTER -> delay variation
- LTE -> packet cellular
- VoLTE -> voice over LTE

STATUS / UNIT / EVIDENCE LIMIT

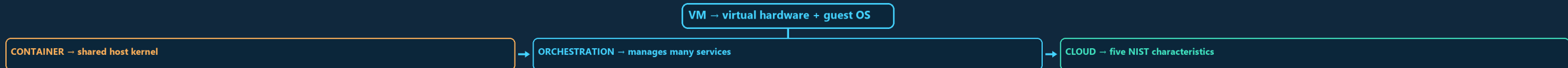
- VLC -> distinct roles
- WIRELESS LABEL -> no universal range or wall penetration

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: WIRELESS LABEL -> no universal range or wall penetration.

08

STAGE 08 VIRTUALISATION AND CLOUD BOUNDARY

VIRTUALISATION AND CLOUD BOUNDARY VIRTUAL HARDWARE + GUEST OS SHARED HOST KERNEL MANAGES MANY SERVICES FIVE NIST CHARACTERISTICS RESPONSIBILITY SHIFTS



SYSTEM / DEVICE / INSTITUTION

- VM -> virtual hardware + guest OS
- CONTAINER -> shared host kernel

DECISIVE TECHNICAL DISTINCTION

- ORCHESTRATION -> manages many services
- CLOUD -> five NIST characteristics

STATUS / UNIT / EVIDENCE LIMIT

- SaaS -> responsibility shifts
- VM -> virtual hardware + guest OS

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: IaaS | PaaS | SaaS -> responsibility shifts.

09

STAGE 09 DATA AND DISTRIBUTED TRADE-OFF

DATA AND DISTRIBUTED TRADE-OFF ORGANISED MANAGED DATA TRANSACTION PROPERTIES AVAILABILITY/LOCALITY, NOT BACKUP CAP DURING PARTITION CONSISTENCY OR AVAILABILITY TRADE PLACEMENT BY LATENCY AND RESILIENCE MUST REMEMBER

SYSTEM / DEVICE / INSTITUTION	DECISIVE TECHNICAL DISTINCTION	STATUS / UNIT / EVIDENCE LIMIT
DATABASE / DBMS -> organised managed data	REPLICATION -> availability/locality, not backup	EDGE / FOG / CLOUD -> placement by latency and resilience
ACID -> transaction properties	CAP DURING PARTITION -> consistency or availability trade	MUST REMEMBER: Computing separates hardware, firmware, operating system, application and...

SYSTEM / DEVICE / INSTITUTION

- DATABASE / DBMS -> organised managed data
- ACID -> transaction properties

DECISIVE TECHNICAL DISTINCTION

- REPLICATION -> availability/locality, not backup
- CAP DURING PARTITION -> consistency or availability trade

STATUS / UNIT / EVIDENCE LIMIT

- EDGE / FOG / CLOUD -> placement by latency and resilience
- MUST REMEMBER: Computing separates hardware, firmware, operating system, application and...

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: MUST REMEMBER: Computing separates hardware, firmware, operating system, application and...

09

STAGE 09

DATA AND DISTRIBUTED TRADE-OFF

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SYSTEM / DEVICE / INSTITUTION

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DECISIVE TECHNICAL DISTINCTION

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STATUS / UNIT / EVIDENCE LIMIT

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- MUST REMEMBER: Computing separates hardware, firmware, operating system, application and...

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: MUST REMEMBER: Computing separates hardware, firmware, operating system, application and.

10

STAGE 10

LEDGER CLAIM FIREWALL

LEDGER CLAIM FIREWALLDISTRIBUTED LEDGERREPLICATED RECORDLINKED BLOCKS + CONSENSUSACCESS MODELSORACLE INPUTCAN BE FALSECONDITIONAL CLAIMS

DISTRIBUTED LEDGER → replicated record

BLOCKCHAIN → linked blocks + consensus

CONSORTIUM → access models

ORACLE INPUT → can be false

SYSTEM / DEVICE / INSTITUTION	DECISIVE TECHNICAL DISTINCTION	STATUS / UNIT / EVIDENCE LIMIT
• DISTRIBUTED LEDGER → replicated record • BLOCKCHAIN → linked blocks + consensus	• CONSORTIUM → access models • ORACLE INPUT → can be false	• NFT / WEB3 / IMMUTABILITY → conditional claims • CLOSE DISTINCTION: Bit is not byte, RAM is not storage, core count is not performance,...

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: CLOSE DISTINCTION: Bit is not byte, RAM is not storage, core count is not performance,.

11

STAGE 11

ADVANCED COMPUTING AND STATUS RAIL

ADVANCED COMPUTING AND STATUS RAILPROCESSOR + ACCELERATOR + INTERCONNECT + SOFTWARESOFTWARE CAPABILITYDEVICE BASEQUBITS FOR SELECTED PROBLEMSCONFIDENTIALITY + INTEGRITY + AVAILABILITYVERIFIED OUTCOME EVIDENCE LIMIT

HPC → processor + accelerator + interconnect + software

AI → software capability

SEMICONDUCTOR → device base

QUANTUM → qubits for selected problems

SECURITY → confidentiality + integrity + availability

CONCEPT / ARCHITECTURE	MECHANISM / APPLICATION	READINESS / STATUS LIMIT
• HPC → processor + accelerator + interconnect + software • AI → software capability • SEMICONDUCTOR → device base	• QUANTUM → qubits for selected problems • SECURITY → confidentiality + integrity + availability • STANDARD → DEMO → DEPLOYMENT → VERIFIED OUTCOME	• EVIDENCE LIMIT: MECHANISM / STATUS / CAUSATION: State layer, component, protocol,...

MECHANISM / VERDICT — EVIDENCE LIMIT: MECHANISM / STATUS / CAUSATION: State layer, component, protocol,...

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: EVIDENCE LIMIT: MECHANISM / STATUS / CAUSATION: State layer, component, protocol,.

E

EXTRA

SUBORDINATE ENRICHMENT — ONLY AFTER THE COMPLETE CORE

OPTIONAL ADVANCED ENRICHMENTMULTIPLE TASKS MAKE PROGRESS DURING OVERLAPPING TIME.TASKS EXECUTE SIMULTANEOUSLY.RACE CONDITIONRESULT DEPENDS ON UNCONTROLLED EXECUTION ORDER.TASKS WAIT INDEFINITELY FOR RESOURCES HELD BY ONE ANOTHER.VIRTUAL MEMORY

MAPS PROCESS ADDRESSES TO PHYSICAL MEMORY/STORAGE, ENABLING ISOLATION AND

OPTIONAL SCIENTIFIC / ENGINEERING DEPTH	READINESS, UNIT OR STATUS LIMIT	COMPARATIVE TECHNOLOGY / GOVERNANCE NUANCE
• Concurrency: multiple tasks make progress during overlapping time. • Parallelism: tasks execute simultaneously. • Race condition: result depends on uncontrolled execution order. • Deadlock: tasks wait indefinitely for resources held by one another.	• Virtual memory: maps process addresses to physical memory/storage, enabling isolation and a • Routing: selecting paths between networks. • NAT: translates address information, commonly allowing many private devices to share a public • DHCP: automatically supplies network configuration.	• Content Delivery Network: distributes cached content closer to users. • Load balancer: spreads requests across servers. • Zero trust: does not grant implicit trust merely because a user/device is inside a network; • Cloud centralises compute and economies of scale.

OPTIONAL STATUS — This optional depth is unnecessary for a competent core answer; use it only for added nuance after the complete core has been written.